COURSE INFORMATION

School/Faculty:	**Faculty of Computing**		
Program name:	**Bachelor in Software Engineering with Honors**		
Course code:	**SCSE2233**	**Academic Session/Semester:**	**20252026/2**
Course name:	**Software Design and Architecture**	**Pre/co requisite (course name and code, if applicable):**	**SCSE1203**
Credit hours:	**3**		

Course synopsis	This course provides the students with an in-depth look at the theory and practice of software architecture and design. It introduces the important concepts related to software architecture and design. It emphasizes on the design and (faithful) implementation of a large scale software using the widely accepted architecture styles and design patterns. It will also expose students to the use of the industrial strength design notations (e.g. UML) and CASE tools (e.g. Ent Arch, Visual Studio). In addition, it provides other aspects of a large and complex software design such as user interface design, management, leadership, and ethics. At the end of this course, the students should be able to use the techniques, architectural styles, and design patterns in software design.
Course coordinator (if applicable)	Dr. Muhammad Khatibsyarbini

Course lecturer(s)	**Name**	**Office**	**Contact no.**	**E-mail**
	Dr. Muhammad Khatibsyarbini	N28	60 13-509 9215	khatibsyarbini @utm.my
	Dr Adila Firdaus Binti Arbain	N28a	60 17-448 2905	adilafirdaus @utm.my
	Prof Dr Wan Nasir b. Wan Abd Kadir	N28a	60 19-771 2410	wnasir@utm.my

Mapping of the Course Learning Outcomes (CLO) to the Programme Learning Outcomes (PLO), Teaching & Learning (T&L) methods and Assessment methods:

No.	CLO	PLO (MQF Cluster Code)	Taxonomies and generic skills	T&L methods	Assessment methods
CLO1	Classify and evaluate proficiently suitable software design solutions based on the investigation on the given functional and non-functional requirements.	PLO2 (C2)	C5 (SC2)	Lecture, active-learning	Asg, T, F, GP, Q
CLO2	Design and construct project design architecture, based on selected architecture style correctly.	PLO3 (C3A)	C5 (SC3)	Project-based learning	Asg, T, F, GP, Q
CLO3	Demonstrate rapport, interact and work effectively with others as both a team member and a leader.	PLO8 (C3F)	P5 A(CC1)	Project-based learning	GP, Pr
Asg – Assignment; T - Test; GP – Group Project; Pr – Presentation; F - FinalExam; Q - Quiz;					

MQF Cluster Code
C1 = Knowledge & Understanding, **C2** = Cognitive Skills, **C3A** = Practical Skills, **C3B** = Interpersonal Skills, **C3C**= Communication Skills, **C3D** = Digital Skills, **C3E** = Numeracy Skills, **C3F** = Leadership, Autonomy & Responsibility, **C4A** = Personal Skills, **C4B** = Enterpreneurial Skills, **C5** = Ethics & Professionalism

Details on Innovative T&L practices:

No.	Type	Implementation

1	Active Learning	Conducted through in-class activities such as reflection, brainstorming, closure pairs review and discussion.

Transferable skills (generic skills learning in course of study which can be useful and utilised in other settings) :
Leadership skill

Student learning time (SLT) / Effective Learning Time (ELT) details:

Week/ Meeting	Course Content Outline and Subtopics	CLO	Learning and Teaching Activities									TOTAL SLT
			Face-to-Face (F2F)								Non F2F Independent Learning	
			Physical				Online (Synchronous)				Online (Asynchronous)	Others
			L	T	P	O	L	T	P	O		
Meeting 1	**Introduction Software Architecture and Design** - Software Design Challenge - Software Design Process - Roles of Software Designer - Software Design Fundamentals	CLO1					2	1			2	2	7
Meeting 2	**UML Revisit** - Introduction to UML diagram - UML Behavioral Model (Use Case Diagram, Sequence Diagram) - UML Structural Model (Component Diagram, Class Diagram) - Visual Paradigm Lab	CLO1 CLO2					2	1			2	2	7
Meeting 3		CLO1 CLO2					1	1	1		2	2	7
Meeting 4	**Principle Software Architecture** - Fundamentals of Software Architecture - Task in Software Architecture - Software Architecture Process - Software Architecture 4+1 view	CLO1	2	1							2	2	7
Meeting 5	**Architecture Style** - Classification of Architecture Styles - Types of Systems and its Architecture Style	CLO2	2	1							2	2	7
Meeting 6	**Principles of Detailed Design** - Task in Detailed Design - Component Design - Open Closed Principle - Liskov Substitution Principle - Interface Segregation Principle - Classification of Design Patterns - Documenting Design Patterns	CLO1 CLO2	2	1							2	2	7
Meeting 7	**Application of Architectural Style and Detail Design** - Documenting Architectural Style - VP Lab Essential	CLO2 CLO3	1	1	1						2	2	7
Meeting 8	**Creational Design Patterns** - Singleton, Abstract Factory, Builder, Prototype	CLO1 CLO2	2	1							2	2	7
Meeting 9	**Structural and Behavioural Design Patterns** - Adaptor, Composite, Façade, Iterator, Observer	CLO1 CLO2	2	1							2	2	7

Meeting 10	**Application of Design Patterns** - Documenting Design Patterns - VP Lab Essential	CLO2 CLO3	1	1	1						2	2	7
Meeting 11	**Software Design Management,Leadership, and Ethics** - Design Management Framework - Planning Management - Leadership and Ethics in Design Management - Work Breakdown Structure - EVM	CLO1 CLO2	2	1							2	2	7
Meeting 12	**Application of Design Document** - SDD Formatting Document - Lab Essential	CLO2 CLO3					1		2		2	2	7
Meeting 13	**Principles of Construction Design** - Algorithmic Viewpoint: Flow-based,state-based, table-based - Construction programming style - Quality Evaluation of Construction - Mc Cabe's Complexity Theory	CLO1 CLO2	2	1							2	2	7
Meeting 14	**Application of Construction Design** - Documenting Constuction Design - Lab Essential	CLO2 CLO3						1	2		2	2	7
											SUB-TOTAL SLT :		**98**

Continuous Assessment		**%**			**Face-to-Face (F2F)**		**NF2F Independent Learning for Assessment**		**TOTAL SLT**
		CLO1	**CLO2**	**CLO3**	**Physical**	**Online (Synchronous)**	**Online (Asynchronous)**	**Others**	
1	Quiz-1	4				0.5			0.5
2	Quiz-2	4				0.5			0.5
3	Assignment-1	2.5	4	2.5	1		2	1	4
4	Assignment-2	2.5	4	2.5	1		2	1	4
5	Project		14	5	1		3	1	5
6	Leadership and Teamwork (3)			10	1		1	1	3
7	Mid Test	9	6		2				2
							SUB-TOTAL SLT :		**19**

Summative Assessment		**%**			**Face-to-Face (F2F)**		**NF2F Independent Learning for Assessment**		**TOTAL SLT**
		CLO1	**CLO2**	**CLO3**	**Physical**	**Online (Synchronous)**	**Online (Asynchronous)**	**Others**	
1	Final Exam	18	12		3				3
							SUB-TOTAL SLT :		**3**
							SLT for Assessment:		**22**
							GRAND TOTAL SLT:		**120**

A	% SLT for F2F Physical Component	30.00
B	% SLT for Online & Independent Learning Component :	70.00
C	%SLT for Online Component:	43.33
D	% SLT for All Practical Component:	5.83
D1	% SLT for F2F Physical Practical Component:	1.67
D2	% SLT for F2F Online Practical Component:	4.17
Please tick (/) if this course is Industrial Training/ Clinical Placement/ Practicum using 50% of Effective Learning Time (ELT)		

*** To pass this course, students must pass both Continuous and Final assessment**

Identify special requirement or resources to deliver the course (e.g.,software,nursery, computer lab, simulation room etc)

References (include required and further readings, and should be the most current)
Text book (if applicable) - M. Richards, Fundamentals of Software Architecture: An Engineering Approach, 1st. Edition, O'Reilly Media, 2020 Main references - V. Sarkar, Design Pattern in C#: A Hands-On Guide with Real-World Examples., 2nd. Edition, Apress, 2020. - Richard N.Taylor, Nenad Medvidovic and Eric M.Dashofy, Software Architecture: Foundations, Theory, and Practice, Wiley, 2009

Other additional information (if applicable)
Academic honesty and plagiarism: *(Below is just a sample)*
Assignments are individual tasks. Copying of work (texts, simulation results etc.) from other students or from other sources is not allowed. Brief quotations are allowed and then only if indicated as such. Existing texts should be reformulated with your own words used to explain what you have read. It is not acceptable to retype existing texts and just acknowledge the source as a reference. Be warned: students who submit copied work will obtain a mark of zero for the assignment and disciplinary steps may be taken by the Faculty. It is also unacceptable to do somebody else's work, to lend your work to them or to make your work available to them to copy.

Other additional information (if applicable)

Attendance is compulsory and will be taken in every lecture session. Student with less than 80% of total attendance is not allowed to sit for final exam.
Students are required to behave and follow the University's dressing regulation and etiquette all the time.
Exercises and tutorial will be given in class and some may be taken for assessment. Students who do not do the exercise will lose the coursework marks for the exercise.
Assignments must be submitted on the due dates. Some points will be deducted for late submissions.
Assignments submitted three days after the due date will not be accepted.
Make up exam will not be given, except to students who are sick and submit medical certificate which is confirmed by UTM panel doctors. Make up exam can only be given within one week of the initial date of exam.

To pass this course, students must pass both Continuous and Final assessment.
All students must adhere to the following to pass a computing course:
i. Pass both continuous assessment AND final assessment
ii. Get at least minimum pass marks following UTM regulation (40%)
- The passing mark for Continuous Assessment is 30% of continuous marks.
- The passing mark for the Final Assessment is 20% of the final marks.-
- If a student does not achieve the passing mark for continuous assessment or the passing mark for the final assessment, then the student's overall final mark will be set below the UTM passing mark.

Disclaimer:

All teaching and learning materials associated with this course are for personal use only. The materials are intended for educational purposes only. Reproduction of the materials in any form for any purposes other than what it is intended for is prohibited.
While every effort has been made to ensure the accuracy of the information supplied herein, Universiti Teknologi Malaysia cannot be held responsible for any errors or omissions.

Prepared by:	**Certified by:**
Name: Dr. Muhammad Khatibsyarbini	**Name:**
Signature: 	**Signature:**
Date: 4 March 2026	**Date:**